

CHAPTER 1

Introduction to the Internet of Things

This chapter introduces what the Internet of Things (IoT) is. To this end, it uses several definitions that have been provided by different organizations. The chapter illustrates the main concepts behind the IoT computing paradigm, along with its main drivers. Moreover, it presents some of the most popular IoT technologies and applications, including the ones that will be detailed in subsequent chapters. At the end of this chapter, examples of popular IoT devices are given. Specifically, devices commonly used in enterprise and educational settings are presented. These devices provide some concrete examples of what “things” look like in the scope of the IoT paradigm. Using these popular devices, researchers, developers, and users can do practical work with IoT and hence understand how IoT applications work in practice.

1.1 Defining IoT

IoT is not tied to a single technology or type of application. Hence, it has been always difficult to provide a precise description of it. There have been several definitions of the IoT, as several organizations have attempted to describe this novel computing paradigm. For example:

- One of the first definitions of IoT was given almost 20 years ago by Kevin Ashton, cofounder of the Auto-ID Center at the Massachusetts Institute of Technology (MIT): “System where the Internet is connected to the physical world via ubiquitous sensors.” This early definition has been used by several organizations including standards development bodies such as the Organization for the Advancement of Structured Information Standards (OASIS).
- The International Telecommunication Union (ITU) defined IoT as [1]: “a global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies.”
- The European Technology Platform on Smart Systems Integration (ETP EPoSS) has defined IoT as: “Things having identities and virtual personalities operating in smart spaces using intelligent interfaces to connect and

communicate within social, environmental, and user contexts.” (https://iot.ieee.org/images/files/pdf/IEEE_IoT_Towards_Definition_Internet_of_Things_Issue1_14MAY15.pdf)

- These definitions reveal the complexity of IoT computing. In general, one can consider IoT as the computing paradigm that enables the combination of data and services from all the entities that are connected to the internet (i.e., all internet-connected objects) towards providing added-value services to citizens and businesses [2]. This definition will become clearer in the rest of the chapter, where we explain the main drivers behind the emergence and rise of the IoT.

Based on these definitions, one may wonder whether IoT is simply a rebranding of older concepts such as ubiquitous and pervasive computing. Indeed, nearly 30 years ago, Mark Weisser at the Computer Science Lab, Xerox PARC, identified three eras of computing:

- *Mainframe computers*: In this era, one (super) computer was used to serve multiple people through their terminal devices. This was typically how computers worked in the 1980s.
- *Personal computing*: In this era, every person used his or her own (desktop or laptop) personal computer.
- *Ubiquitous computing*: In this era, many computers provide services to one person. This is practically the present era, as it is now common for people to possess many devices (e.g., laptops, mobile phones, tablet computers). Moreover, most computing services are delivered based on more than one computing system including different types of pervasive computing systems such as smart sensors, robots, and wearable devices.

Overall, the vision of ubiquitous computing, as expressed three decades ago, included several of the concepts of the IoT computing paradigm. However, in the past decade, there has been a surge of interest in IoT for various reasons, as explained next.

1.2 Why IoT Now?

Several factors have driven the emergence and rise of IoT. Several of them have appeared intensified in the past few years, and this has significantly increased the relevance of the IoT paradigm.

1.2.1 Technology Acceleration

We are living in the era of technology acceleration, where the pace of technological progress speeds up exponentially over time. This is particularly true for information technology, which is growing exponentially and propelling accelerated change as

well. The technological acceleration is reflected on several factors that boost IoT's growth.

1.2.1.1 Proliferating Number of Devices

As part of the technological acceleration, the number of internet-connected devices is proliferating in an exponential pace. According to the IoT Analytics research firm, the number of internet-connected devices on the planet in 2019 exceeded 7 billion. This number did not account for devices such as tablets and smartphones. When considering smartphones and portable computers, the number of connected devices exceeds 17 billion. Furthermore, it is expected that the number of connected devices will continue to grow exponentially. According to recent research by Transforma Insights, the number of active IoT devices is expected to exceed 24 billion in 2030 [3]. The same study indicated a compound annual growth rate (CAGR) of 11% in the number of IoT devices. It should be also noted that the number of connected devices increases at a greater pace than the number of people on the planet. It was around 2005 when the number of connected smart phones exceeded the planet's population. In 2008, the number of IoT devices exceeded the number of people on the planet, even without counting smart phones and laptops. The point in time where the number of internet-connected devices exceeded the number of people on the planet is considered a major milestone in the evolution of the IoT paradigm.

1.2.1.2 Different Types of Devices

In addition to the exponential increase in the number of devices, we are also witnessing a proliferation of the different types of connected devices. Smart watches, Fitbits, smart textiles (e.g., clothing that can sense temperature and other parameters), boards that can host multiple sensors and actuators, connected thermometers, smart energy meters, internet-connected cameras, and pervasive sensors that can be attached to sports shoes are only some of the available internet-connected devices. These devices serve different purposes, measure different parameters, and are deployed in various IoT applications. It is nowadays common for consumers to purchase multiple devices of different types. At the same time, thousands of different devices are also deployed in industrial settings such as manufacturing shop floors, energy plants, and oil refineries.

Another factor that justifies the emergence and rise of IoT is that all these different devices are connected to the Internet. This is no longer the norm just for computers and smart phones, but also for a larger number of other devices such as coffee machines, cars, and electronic scales.

1.2.1.3 Moore's Law and the Exponential Increase in Computing Power

In 1965, Gordon Moore, the chief executive officer (CEO) of Intel, observed that the number of transistors in a dense integrated circuit doubles about every 2 years. At that time, nobody expected that this observation would go on for decades. Moore projected that this rate of growth would continue for at least another

decade. However, this exponential growth has been continuing to date, which facilitates the fast processing of large amounts of IoT data. The abundance of computing cycles is therefore one more factor that contributes to the rapid growth of IoT.

1.2.2 People Connect to Things

Another growth factor for IoT is that people can connect to things. As things are connecting to the internet, they can access information systems (e.g., a hospital information system or an enterprise resource planning (ERP) system), but also user devices (e.g., wearables and smart phones). Therefore, user devices enable human users to interact with other devices and facilitate things' interactions with people. As a prominent example, a person instrumented with sensors (e.g., motion sensors and electrocardiogram (ECG) sensors) can interact with a smart phone. Different interaction scenarios between a person and a device can be implemented in this context.

1.2.3 Things Connect to Things

Another IoT interaction scenario involves things connecting to things for the purpose of exchanging information or even enabling one system (thing) to invoke and take advantage of the services of the other (thing). Interactions between things are very common in the IoT paradigm. As a prominent example, a vehicle can interact with other vehicles as part of vehicle-to-vehicle (V2V) applications. Likewise, a vehicle can interact with the infrastructure of the city, which is usually referred to as vehicle-to-infrastructure (V2I). V2V and V2I are among the most common IoT applications in smart transport. The interactions between the various things are enabled by complex and heterogeneous networks that facilitate the connectivity of the various devices [4].

1.2.4 Climate Change

Climate change is one of the most important phenomena of our age. It refers to a long-term shift in average weather patterns, which results in significant and abnormal variations in weather conditions. To fight climate change, governments and enterprises all around the world design and implement “green” projects that improve environmental performance. Several IoT applications hold the promise to boost environmental performance and green friendliness, such as projects that boost the greater use of renewables. Such projects increase the demand for IoT applications and propel IoT's growth.

1.2.5 The Rise of the Fourth Industrial Revolution (Industry 4.0)

Technological revolutions are typically characterized by the rapid introduction of new technologies that replace legacy ones in short amounts of time. An industrial

revolution is taking place during the past few years, based on the introduction of IoT, artificial intelligence (AI) and other digital technologies [5]. These technologies enable the digitalization of physical processes and drive rapid change in industrial sectors such as manufacturing, energy, oil and gas, mining, and healthcare. This industrial revolution is characterized as the fourth industrial revolution (Industry 4.0). Industry 4.0 blurs the boundaries between the digital, physical, and biological worlds. IoT is playing a key role in the fourth industrial revolution, which also explains the surge of interest in IoT technologies and applications.

1.3 IoT Versus Related Technologies

IoT is very closely related to various other technological concepts, which involve the integration and coordination of data and services from multiple connected devices. Some of these concepts are used interchangeably with IoT. However, in most cases, these technological terms and buzzwords have either subtle or bigger differences to the IoT paradigm. Let us explore some of the terms that are very relevant to IoT and are, in several cases, used instead of it.

1.3.1 Ubiquitous Sensor Networks (USN)

USN refers to applications that are empowered by large sets of connected sensors, which are, in most cases, distributed both geographically and administratively. USN can be considered as a subset of IoT in the sense that it considers the interconnection and coordination of multiple sensing devices. However, IoT is not limited to sensing devices and therefore IoT has a much broader scope than USN.

1.3.2 Machine-to-Machine (M2M) Communications

M2M is a very commonly used term among telecommunications organizations, which use it to denote things-to-things interactions. As already outlined, M2M or things-to-things interactions are supported by IoT. However, IoT supports additional interactions as well and hence it is a superset of M2M. Nevertheless, M2M applications are certainly a very important segment of IoT.

1.3.3 Internet of Everything (IoE)

IoE is another term used instead of IoT. It has been propelled by IoT vendors such as Cisco. IoE refers to the networked connection of things, people, data, and processes, and, as such, it is sometimes considered broader than IoT. Nevertheless, the scope of IoT includes the connection of devices, people, and processes as well. Hence, IoT can be used interchangeably with IoE. The choice between the two terms is usually based on marketing and branding criteria, rather on the different capabilities and functionalities implied by each one of the two.

1.3.4 Cloud of Things (CoT)

CoT is common in cases where IoT data and services are deployed and/or used in the cloud. It is a term inspired by the very close affiliation between IoT and cloud computing. Nevertheless, IoT applications are not, in all cases, cloud-based, which makes CoT a subset of IoT.

1.3.5 Web of Things (WoT)

WoT refers to systems that converge web concepts with IoT concepts towards enabling end users or developers to access IoT data and IoT services over the web [6, 7]. The emergence of the WoT concept has its roots in that many IoT applications are web-based. It should be noted that not all IoT applications are web-based, which makes WoT a subset of IoT rather than a synonym for it.

1.3.6 Cyber-Physical Systems (CPS)

CPS refers to networked systems that combine information and services from both the physical world and the cyber world. Moreover, CPS use information from the digital world to drive physical processes and the operation of physical parts of the system. It is therefore evident that CPS functionality is covered by IoT definitions. In practice, CPS is used instead of IoT in cases of systems that comprise physical systems and processes, while at the same time involving actuation and control. This is the case in several application areas such as smart manufacturing in the scope of the fourth industrial revolution (Industry 4.0), which is enabled by CPS.

One will certainly listen, read, and use some of these terms instead of IoT. It is good to use the proper term in the proper context, which requires an understanding of the scope and the meaning of each of these terms.

1.4 IoT Technologies and Applications

1.4.1 IoT Technologies

IoT is not a single technology, which is why its definition is quite complex and comes in many different flavors. Rather, IoT is an entire paradigm that is empowered by a pool of complementary technologies such as sensors, wireless sensor networks (WSNs), radio frequency identification (RFID), semantic technologies, cloud computing, edge computing, big data, AI, blockchains, cybersecurity, and augmented reality (AR). These technologies (e.g., cloud computing and big data) are the digital enablers of the IoT paradigm. Each of them plays a role in the development, deployment, and operation of IoT systems and applications. The rest of this book illustrates many of these technologies and the ways in which they are affiliated with IoT, but also the ways in which they are used in IoT applications.

Given that IoT deployments are not based on a single technology but rather on the integration of multiple technologies, nontrivial IoT systems comprise various components such as:

- *Sensors and actuators*: These provide the means for acquiring information from the physical environment (i.e., sensors) and providing actionable commands to devices and physical processes (actuators). Sensors and actuators enable information technology (IT) systems to interact with the physical world (i.e., the field).
- *Communication components*: These enable the interactions between the distributed nodes of an IoT system.
- *Middleware platforms*: These provide the digital plumbing between devices (such as sensors) and the end-user applications.
- *Data analytics engines*: These undertake the processing of IoT data towards converting raw IoT information to knowledge.
- *Applications*: These provide the user interface (UI) that enables end users to interact with IoT applications.

This list of components is nonexhaustive. In subsequent chapters, we will provide an anatomy of IoT applications through an in-depth presentation of some of these components.

For example, WSN is a special element of IoT systems, which comprises a set of sensors that are configured to form a network. This network of sensors ends up connected to a sink node, which conveys information from the network to other internet-connected entities and devices. WSNs can be considered a component of IoT systems, while at the same time being themselves a special class of IoT applications. WSN was one of the earliest instances of IoT deployments and is, in several cases, considered a forerunner of IoT. Chapter 2 is devoted to the presentation of WSN technologies.

As another example, IoT is very closely affiliated to cloud computing technologies. Sophisticated IoT systems can greatly benefit from the scalability, capacity, and pay-as-you-go characteristics of the cloud. As modern IoT systems deal with large amounts and data, cloud computing provides a compelling value proposition for hosting IoT data and executing IoT services. Moreover, the cloud enables companies to gradually scale up their deployments. As a result, IoT is, in several cases, studied and considered in conjunction with cloud computing. Likewise, most IoT architectures make provisions for the integration of IoT with the cloud. Specifically, the cloud acts as an intermediary between the internet-connected devices and the end-user applications in the various IoT applications areas. It offers services such as storage, computation, and visualization based on popular cloud computing models such as Software-as-a-Service (SaaS), Infrastructure-as-a-Service (IaaS), and Platform-as-a-Service (PaaS). At the same time, there are new cloud computing models for IoT. Chapter 4 is dedicated to the discussion of IoT and cloud integration, where such novel IoT-based cloud computing models are presented.

1.4.2 IoT Applications

There is no single IoT application type but rather a host of different areas where IoT can be applied [8, 9]. Prominent examples include IoT's applications in:

- *Healthcare*: Here IoT opens, for example, new horizons in disease management through a wide range of devices that continually acquire and process information about the context of the patient.
- *Manufacturing*: Here IoT enables new efficiencies both in factory automation and in the efficiency of the supply chain. Specifically, IoT facilitates the seamless flow of information about products and materials across all stakeholders of the supply chain and their information systems.
- *Transport*: Here IoT is enabling completely new modes of transport such as connected cars and self-driving cars, which will radically improve the safety and economic efficiency of driving.
- *Buildings*: Here IoT-based multisensor systems provide the means for efficient management of the buildings, including, for example, advanced features for energy management and facility management.
- *Smart cities*: These provide the urban development context where all the above listed applications and innovations are executed. IoT-enabled smart cities provide advanced functionalities that respond to the challenges of urbanization through alleviating resource depletion and overall improving quality of life. To this end, they integrate a wide range of other IoT-enabled applications such as applications for aging well and efficient urban mobility.

As shown in Figure 1.1, different IoT applications can be developed over a pool of IoT infrastructures that comprise networks, sensors, connected devices, and data. To this end, application developers leverage middleware services provided by IoT infrastructures and middleware.

IoT is completely transforming some of these areas through enabling new disruptive products and services that significantly enhance contemporary service offerings. Chapter 10 presents IoT-enabled applications in the above areas. However,

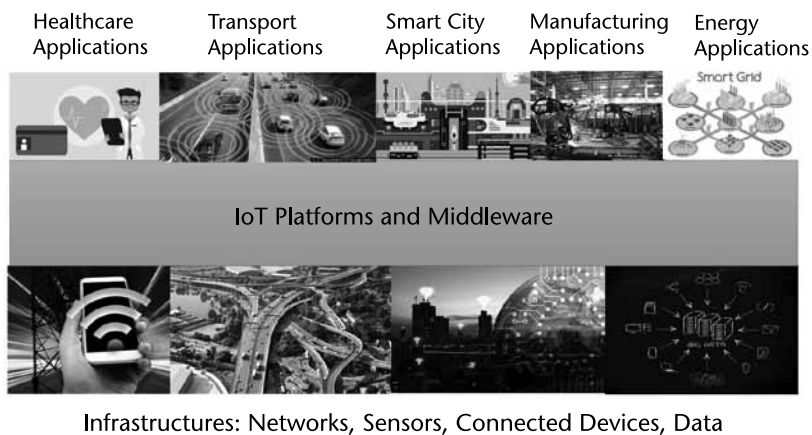


Figure 1.1 IoT platforms enable the development of different types of IoT applications over networks, sensing, and data infrastructures.

this list of IoT applications is not exhaustive, as the scope of IoT deployments is expanding every day to more application areas.

In some cases, IoT applications are also classified based on their spatial and geographical scope. For example, there are applications for:

- *The home environment (smart home applications)*: Such as applications that manage in-home utilities and resources;
- *Communities (smart communities)*: Such as applications for renewables and smart energy in urban environments;
- *National scale deployments*: Such as natural resources optimization applications;
- *Transport and mobility applications*: These provide support to users while they are on the move.

These applications are connected to each other through IoT infrastructures, including networks and cloud computing infrastructures. In general, there are many different classifications of IoT applications based on different criteria such as the type of the application, its geographical scope (see Figure 1.2), and its timescale of execution (e.g., real-time or non-real-time).

1.5 Reference Models and Architectures for IoT Systems

In order to understand the elements of an IoT system, one can study reference architecture models, which identify the components that typically comprise an IoT system. Some of the most prominent reference models for IoT systems are presented next.

1.5.1 Industrial Internet Consortium Reference Architecture (IIRA)

The IIRA specifies a common architecture framework for developing interoperable IoT systems for different vertical industries. It is an open, standards-based architecture, which has broad applicability. The latter makes it a vehicle for interoperability, mapping, and practical deployment of IoT technologies, as well as standards

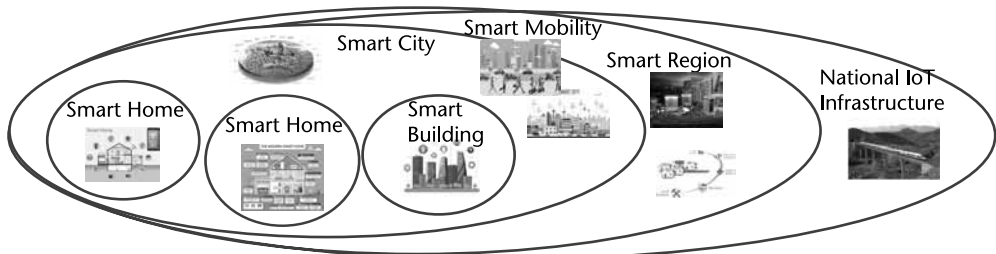


Figure 1.2 The different geographical scope of IoT applications.

development. As a result of its broad applicability, the IIRA is general, abstract, and high-level. Hence, it can be used to drive the structuring principles of an IoT system, without specifying its low-level implementation details. It is also a very good vehicle for communicating concepts and facilitating stakeholders' collaboration.

Based on the analysis of multiple use cases in different sectors, the IIRA presents the structure of IoT systems from four viewpoints, namely, business, usage, functional, and implementation viewpoints. Among these four viewpoints, it is the functional viewpoint that specifies the functionalities of an IIoT system. To this end, the functional viewpoint specifies distinct functionalities in the form of the functional domains. Functional domains can be used to decompose an IoT system in a set of important building blocks, which are applicable across different vertical domains and applications. As such, functional domains are used to conceptualize concrete functional architectures. The IIRA decomposes a typical IoT/Industrial Internet of Things (IIoT) system into five functional domains, namely a control domain, an operations domain, an information domain, an application domain, and a business domain. The implementation viewpoint of the IIRA is based on a three-tier architecture, which follows the edge or cloud computing paradigm. It includes an edge, a platform, and an enterprise tier, which are among the most common elements of a nontrivial IoT system.

1.5.2 OpenFog Reference Architecture

The OpenFog Consortium was a consortium of high-tech industrial enterprises companies and research or academic institutions, which are collaborating towards standardizing and promoting the fog computing paradigm. Fog computing is directly associated with IoT, as it leverages fog nodes (i.e., IoT devices and gateways) in order to enable reliable, low-latency IoT applications. Fog computing alleviates the limitations and drawbacks of conventional cloud computing in various scenarios where low-latency and processing close to the field is required. The reference architecture (RA) of the OpenFog Consortium illustrates the structure of fog computing systems. It presents how fog nodes can be connected partially or fully to enhance the intelligence and operation of an IoT system. Moreover, it presents solutions about growing system wide intelligence away from low-level processing of raw data. The RA is described in terms of different views, including functional and deployment views. The RA specifies some cross-cutting functionalities, which are applied across all layers of an IoT system. These cross-cutting functionalities are conveniently called perspectives.

In January 2019, the Industrial Internet Consortium (IIC) and the OpenFog Consortium (OpenFog) announced that they have finalized the details to combine the two largest and most influential international consortia in industrial IoT: fog and edge computing. Hence, the OpenFog RA is no longer evolving independently. Rather relevant work is conducted in conjunction with the Industrial Internet Consortium.

1.5.3 IoT Reference Architecture Model

A reference model for IoT systems was also developed in the scope of the EU IoT-A project (www.iot-a.eu). The model is currently maintained by the IoT Forum (<https://iotforum.org/>). IoT-A produced an architectural reference model (ARM) for IoT systems, which addressed interoperability issues that affect nontrivial IoT systems that comprise many heterogeneous components. The IoT-A ARM relies on five different views of submodels.

Specifically, IoT-A comprises a domain model, which defines a set of high-level components or actors that are common to most IoT use cases. Likewise, an information model delves into the analysis, specifying common attributes for each one of the entities of the domain model and their semantics. From the domain model, the IoT-A ARM introduces a set of functionality groups (FGs) as part of the functional model. Specific FGs are included to handle the peculiar characteristics of IoT communication flows. The communication model relies on these components of the architecture and on ISO OSI reference model [10] to address interoperability at the stack or network level. Finally, security aspects are in the scope of a trust, security, and privacy model.

1.5.4 ISO/IEC CD 30141 Internet of Things Reference Architecture (IoT RA)

ISO/IEC CD 30141 Internet of Things Reference Architecture (IoT RA) is an ongoing standardization process. ISO/IEC 30141 performed a heuristic analysis of system characteristics that are common in most IoT systems (e.g., auto-configuration, discoverability, scalability). From them, it proposes a conceptual model where typical IoT entities or actors and their relationships are described. The CM is quite like the IoT-A ARM domain model and includes IoT users, applications, services, or virtual entities. Finally, the five different views that describe the architecture are presented: functional, system, communications, information, and usage.

Other reference models have been produced by vendors, such as Microsoft (e.g., the Azure IoT Reference Architecture [11]) and Amazon [12].

Table 1.1 illustrates provides an overview of the different RAs for IoT Systems. For every RA, it lists the standards development organization (SDO) in charge, the main focus of the architecture, and the target application sectors.

1.6 The Different Layers and Viewpoints of IoT Systems

1.6.1 Layers of IoT Systems

Most of the IoT reference architectures provide layered views of IoT systems [13]. These views provide another good way to understand the various components that comprise and IoT application. While there is no common layered view across the different architecture models, the following layers are in generally found in IoT systems.

Table 1.1 Overview of Different RAs for IoT Systems

<i>Short Name_ Organization in Charge</i>	<i>Type/Emphasis</i>	<i>Target Sector</i>	
IIRA	Industrial Internet Consortium	Framework for interoperable IoT systems	Various vertical industrial sectors (e.g., manufacturing, energy, healthcare)
OpenFog RA	OpenFog Consortium	Structuring principles for fog computing systems	Smart cities and sectors where fog computing is used
IoT-A ARM	IoT Forum	Reference model for creating IoT architectures	Applicable to all sectors
ISO/IEC CD 30141 IoT RA	ISO/IEC	Conceptual models for IoT systems, actors, and their relationships	Applicable to all sectors

1.6.1.1 Sensing and Actuating Layer

This is the layer where various internet-connected devices reside. These include the various sensors and actuators, but also the servers and gateways to which they are typically attached. The term “sensing” is quite limiting in this context, as IoT enables a much broader range of interactions with the physical world, such as intelligent braking systems or socially assistive robots.

1.6.1.2 Network Layer

This layer ensures the connectivity of the internet-connected objects, based on the various communications and networking technologies. Internet-connected objects connect to IoT systems based on different networks, depending on their appropriateness but mainly their availability. For example, several devices (such as smart phones and wearables) connect to the internet through Wi-Fi hotspots. In practice, Wi-Fi is one of the networks that carry most of the IoT traffic worldwide. Cellular phones connect to the internet through mobile networks of various generations, including 3G and 4G/Long Term Evolution (LTE) networks. There are also cases of objects and devices (e.g., devices within ships), which connect to the internet and IoT based on satellite networks. Based on these networking technologies, IoT deployments leverage functionalities such as capacity management, abstraction of resources, and control of operations. Moreover, most of these technologies are based on Internet Protocol (IP)-based backbone networks.

1.6.1.3 Middleware Layer

This layer converts the low-level networked data source to high-level information consumed by IoT applications towards improving business processes and enabling optimized decisions. Typically, this layer includes middleware platforms that process large amounts of data from the lower layers. The processing involves data

filtering and analytics, including functions for mining large datasets such as big data analytics and AI.

1.6.1.4 Application Layer

The applications layer includes the various applications that can access information and services of the internet-connected devices, through the middleware layer. It also involves the proper visualization of data and services based on dashboards, charts, and more advanced data cyber-representations of data such as virtual reality (VR) and augmented reality (AR).

Figure 1.3 illustrates a layered view of IoT systems based on the above-listed layers.

1.6.2 IoT System Perspectives

Many different stakeholders and actors are involved in IoT systems development and deployment. These stakeholders tend to develop different views and perspectives about IoT. For example, some of them focus on the development of sensors and sensing technologies, including intelligent sensors and energy efficient devices. These people have a “things”-oriented view of the IoT, as they focus on IoT devices, such as smart sensors, smart actuators, and robotic devices.

Other actors consider IoT as a pool of internet-based technologies for accessing information and services on devices. They approach IoT from the perspective of the internet and the web and therefore develop an internet-oriented view of IoT.

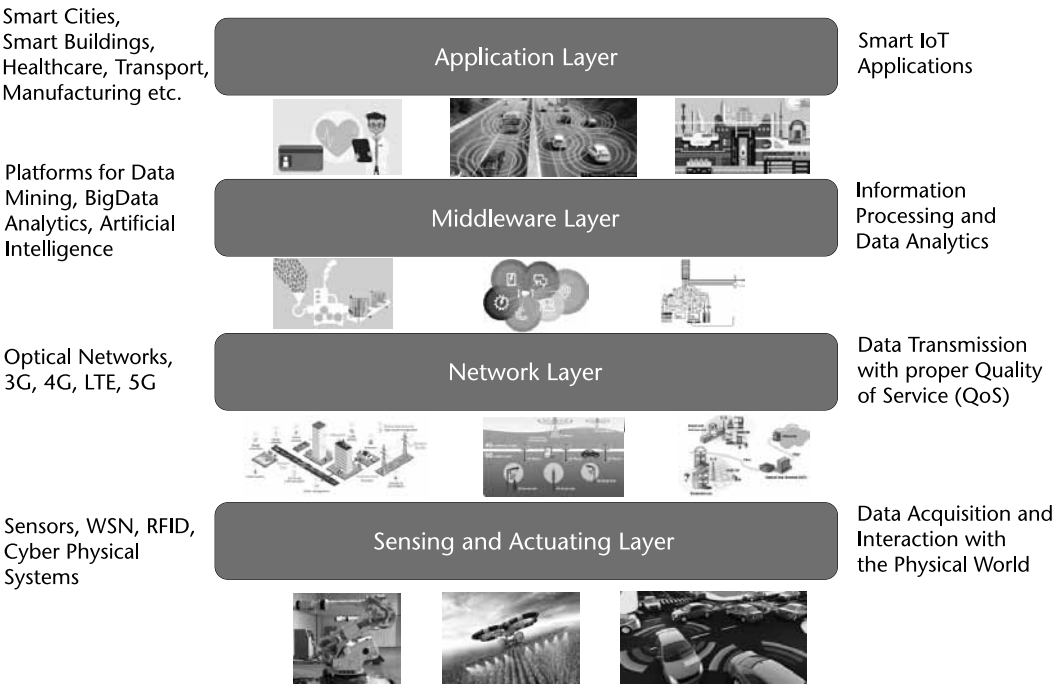


Figure 1.3 Layered structure for IoT systems.

There are also people that work on the semantic unification of the highly heterogeneous landscape of internet-connected objects. They have the semantic-oriented perspective of the IoT.

Other visions and perspectives are also developed. For example, there are actors that perceive IoT as a collection of cloud APIs for application development. Moreover, there are engineers that are completely focused on the networking aspects of IoT. For example, they focus on high-speed connectivity and optimized network quality of service (QoS) for IoT applications.

In practice, IoT is the intersection of all these technologies and visions, especially when it comes to large-scale IoT systems and their integration. Nevertheless, the breadth and depth of IoT technologies make it reasonable for different people to focus on their different areas of expertise. In this book, we try to present many different IoT technologies and applications to provide a 360° view of what IoT is all about.

1.7 Evolution of IoT Systems

For over a decade, IoT systems and applications have been evolving in rapid pace. Along with the proliferation of internet-connected devices, we have also witnessed radical changes in the architectures of IoT systems, as well as in the sophistication of their functionalities. In general, we can distinguish between four phases in the evolution of IoT systems, including (Figure 1.4):

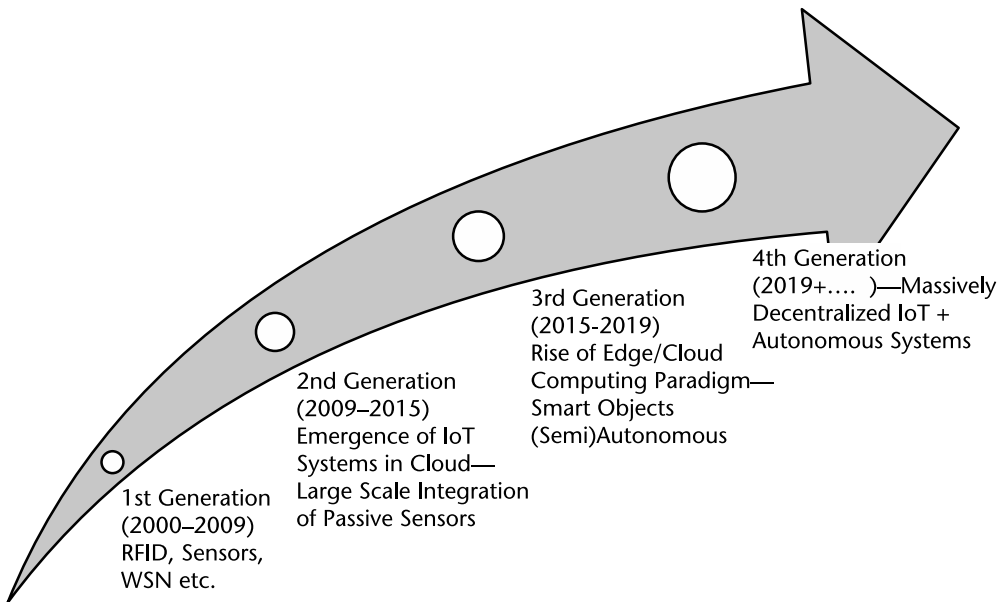


Figure 1.4 Evolution of IoT systems.

- *First generation (2000–2009)*: This phase included the development of IoT systems with sensing functionalities based on technologies such as WSN and RFID. The main characteristics of these systems were that they were passive and mainly focused on the collection, analysis, and visualization of data in a pervasive way.
- *Second generation (2010–2015)*: As part of this phase, early passive and nonscalable IoT deployments were replaced by cloud-based IoT applications, which provided scalability and elasticity in managing large numbers of passive devices (e.g., sensors). This second generation of the IoT system was driven by the emergence and rise of cloud computing.
- *Third generation (2015–2019)*: In this phase, cloud-based IoT systems were gradually decentralized based on the introduction of the edge computing paradigm, which facilitates operations close to the field and offers various performance, scalability, security, and privacy advantages. The notion of computing at the edge of the network had been around several years before. However, it was only in recent years where it was formalized, standardized, and deployed at scale. Note the mainstream reference architectures for IoT systems (such as the IIRA and the OpenFog RA), structure IoT implementations as cloud and edge computing systems.
- *Fourth generation (2019–present)*: In recent years, IoT systems are increasingly comprising smart objects with semiautonomous behaviors, such as autonomous guided vehicles, drones, and smart wearables. These objects operate independently of a single edge or cloud administrative domain and take advantage of recently developed edge computing architectures that support mobility and multiple access points such as multi-access edge computing (MEC). The latter enables a higher decentralization of IoT applications. In the coming years, IoT systems may be transformed towards exhibiting a higher degree of decentralization, beyond a single edge or cloud computing infrastructure. In this direction, decentralization technologies such as blockchains are likely to play a role.

In this book, we present the systems and technologies that comprise the first three generations of IoT systems, while developing a vision for future IoT systems. The latter will be driven by the following trends:

- *Proliferation of semiautonomous and mobile smart objects*: In the coming years, the number of smart, autonomous, and mobile objects will explode. For example, recent projections state that by 2030, there will be 700 million connected cars, 90 million (semi)autonomous vehicles, and 250 million electric and hybrid vehicles on the roads, along with 600 million subscribers to various mobility services.
- *Expanded use of AI*: IoT systems are providing a wealth of data, which propels the deployment of AI systems and applications. AI systems are increasingly deployed and used along with IoT systems in areas like agriculture, healthcare, and industry. AI represents one of the fastest growing markets.

According to Tractica, the global AI software market is expected to grow from approximately \$9.5 billion in 2018 to \$118.6 billion by 2025 [14].

- *Coexistence and collaboration between human and IoT systems:* Despite the rise of robots and automation, humans remain the most flexible and intelligent resource in most IoT applications. In the coming decades, humans are set to coexist and collaborate with IoT systems. Typical scenarios include human-robot collaboration (Cobots) in industrial plants, as well as coexistence of autonomous cars and legacy vehicles in future transport scenarios.

Following chapters explore these trends and how they are likely to impact the future IoT architectures and applications.

1.8 Getting Started with IoT Devices: Sensors and Actuators

The starting point for developers, deployers, and researchers that would like to understand IoT in practice is always sensors and actuators. These are the most fundamental components of an IoT system. They were the main parts of early IoT systems, but they are still an integral element of more sophisticated edge or cloud IoT systems. Hence, they are worth exploring as part of our introduction to IoT systems.

1.8.1 Introduction to Sensors and Actuators

IoT is largely about leveraging data and services from internet-connected devices. Sensors are the most popular IoT devices. A sensor provides usable output in response to a specified measurement. Specifically, it acquires a physical parameter and converts it into a signal suitable for processing (e.g., optical, electrical, mechanical). Sensors are deployed in many different contexts including human bodies, automobiles, airplanes, cellular telephones, radios, chemical plants, and industrial plants. Subsequently, these are deployed and used in many different applications such as healthcare, buildings, transport, energy, water management, and smart cities' applications.

In order to understand the different types of sensors, we present in Table 1.2 different types of stimuli, which are as part of a sensing functions converted to different signals. In particular:

- An acoustic stimulus can produce a wave (amplitude, phase, polarization), spectrum, or velocity related signal.
- A biological and chemical stimulus produces fluid concentrations (i.e., based on gas or liquid).
- An electric stimulus can result in charge, voltage, current, electric Field (amplitude, phase, polarization), conductivity or permittivity signal.
- A magnetic stimulus produces a magnetic field (amplitude, phase, polarization) or flux or permeability signal.

Table 1.2 Sensing Functions

<i>Stimulus Type</i>	<i>Signal/Quantity</i>
Acoustic	Wave (amplitude, phase, polarization), spectrum wave, velocity
Biological and chemical	Fluid concentrations (gas or liquid)
Electric	Charge, voltage, current, electric field (amplitude, phase, polarization), conductivity, permittivity
Magnetic	Magnetic field (amplitude, phase, polarization), flux, permeability
Optical	Refractive index, reflectivity, absorption
Thermal	Temperature, flux, specific heat, thermal conductivity
Mechanical	Position, velocity, acceleration, force, strain, stress, pressure, torque

- An optical stimulus can give a refractive index, reflectivity, or absorption signal.
- A thermal stimulus can result in a temperature, flux, specific heat, or thermal conductivity signal.
- A mechanical stimulus can give a position, velocity, acceleration, force, strain, stress, pressure, and torque signal.

Sensors and actuators can both be characterized as transducers. The role of a transducer is to convert a primary form of energy into a signal with a different energy form. The term “energy form” refers to mechanical, thermal, electromagnetic, optical, chemical, and other forms of energy. Specifically:

- A sensor acquires information from the real world, by detecting and measuring a signal or stimulus. We have already presented different stimuli that are commonly used in IoT applications.
- An actuator generates a signal or stimulus, which it usually provides from the digital world to the real (physical) world.

Both sensors and actuators can be deployed as part of an intelligent feedback system to close its loop of operations: The sensor derives information from the real world and converts it to a digital representation of physical quantities. Likewise, the actuator provides to the physical world (for example, a device or physical process) information or a stimulus or trigger produced in the digital world. This closed loop operation is usually an essential part of a CPS.

Based on the general principles of the sensing paradigm, a wide range of sensors have been developed and are commercially available. Furthermore, they are extensively and increasingly used in several applications. Some prominent examples of sensors and their applications are:

- *Temperature sensors:* Temperature can be measured through pressure, volume, electrical resistance, and strain. Temperature sensing is used in several applications and contexts, including buildings, chemical process plants, engines, appliances, and computers.

- *Accelerometers*: These sensors measure along one axis and are insensitive to orthogonal directions. Accelerometer sensors are commonly integrated in most state-of-the-art smart phones. Beyond integration in smart phones, their applications include vibrations, blasts, impacts, shock waves, air bags, washing machines, heart monitors, and car alarms.
- *Light sensors*: They are composed of a photoconductor, such as a photoresistor, photodiode, or phototransistor. Their applications include cameras, infrared detectors, and ambient lighting applications.
- *Ultrasonic sensors*: They are used for position measurements. In the scope of their operation, sound waves emitted are in the range of 2 to 13 MHz. Two typical examples of ultrasonic sensors include Sound Navigation And Ranging (SONAR) and Radio Detection And Ranging (RADAR) sensors.
- *Photogate sensors*: They are typically used in counting applications (e.g., motion period identification). As part of their operation, they record the time(s) at which light is broken. Photogate sensors include infrared transmitter and receiver at opposite ends of the sensor.
- *CO₂ gas sensors*: They measure gaseous CO₂ levels in an environment. Their measurements are in the range of 0 to 5,000 ppm (parts per million) and are based on the monitoring of infrared radiation absorbed by CO₂ molecules.

1.8.2 Sensor Selection Criteria

The list of presented sensors is obviously nonexhaustive. There are tens of different types of sensors, while new types of innovative sensors are continually emerging. A significant part of IoT innovation lies in the sensing part of IoT systems. Novel sensing techniques and mechanisms are in several cases enabling innovative and disruptive applications. There are numerous examples of novel sensors, such as temperature-sensitive RFID tags (i.e., sensor tags) or ingestible pills that ensure adherence to medication schedules. Depending on the target applications, developers and deployers of IoT applications are offered with many different sensor selection options. However, even when deciding about the type of sensor needed, developers and deployers have many different choices in terms of vendors and qualities. In general, the sensors' selection criteria can be classified in three different categories (i.e., cost-related, environment-related, sensor characteristics-related) as follows:

- *Economic criteria*: Economic criteria include the cost, the availability, and the lifetime of the sensor. These cost-related parameters are part of the technical specifications of a sensor. They can be used (along with the cost of purchase of the sensor) to calculate and appropriately plan the total cost of ownership (TCO) of the sensor. For the TCO calculation, it is not only the absolute cost that matters, but also what will be the duration of use and overall availability of the application based on the sensor.
- *Environmental criteria*: Environmental criteria include the size, the power consumption, and the interference and sensitivity of the sensor. This set of criteria is used with a dual objective:

- To plan and ensure the energy efficiency of the application, including its sustainability;
- To consider the appropriateness of the sensor for the environment at hand, as the sensing capability and accuracy of a sensor vary based on environmental conditions including, for example, the presence of other sensors or electromagnetic devices.
- *Sensor characteristics:* These include the properties of the sensor, including its sensitivity, range, stability, error, and response time. These characteristics should be evaluated against the requirements of the IoT deployment, which is closely related to the business requirements of the target IoT application. These characteristics should be considered to ensure the feasibility of the target deployment. A sensor that does not come with the proper properties and technical characteristics may not be able to support a given application. Beyond feasibility, the various properties should be audited against their ability to optimize the IoT application from a techno-economic perspective (i.e., ensuring both its technical robustness and quality, but also its value for money).

1.8.3 Simple IoT Devices: Arduino Boards and Raspberry Pi

1.8.3.1 Arduino Boards

In most cases IoT devices host multiple sensors and actuators. Consider, for example, the Arduino board [15], which is one of the most popular IoT devices. Arduino is an open-source electronic prototyping platform, which comprises flexible and easy-to-use hardware and software. It is very popular among several groups of IoT applications, including artists, designers, and hobbyists. In the IoT world, it is also extensively used for IoT education and training, as it provides a low-cost and versatile device for explaining what IoT is about. Arduino is appropriate for applications involving interactive objects or environments.

In practice, Arduino is a microcontroller, which can be deployed as a bridge between the cyber and physical worlds. It balances functionality and ease of use, while it can be acquired at a very low cost. Its price starts from as low as \$35. Among the main merits of Arduino is that it is a programmable device through the Arduino C language. The Arduino board comes in various types with different characteristics including Leonard, Due, Micro, LilyPad, Esplora, and Uno.

Among the main reasons for Arduino's popularity is its programmability. Arduino provides the means for programming its operation based on a C++ based language. Arduino programs are called sketches. Each sketch needs at least two functions:

- *void setup ():* This runs first and is run once. It is the place where several initializations take place.
- *void loop ():* This runs over and over, until power is lost or a new sketch is loaded. It hosts the application logic of the board's operation.

Sketches can access the PINs (Pressure Ins) of the board, as PINs are accessible as global variables. Hence, other Arduino programming functions include:

- *digitalWrite(pin, value)*: This sends a voltage level (specified by value) to the designated pin.
- *pinMode(pin, mode)*: This designates the specified pin for input or output (as specified by mode).
- *digitalRead(pin)*: This reads the current voltage level from the designated pin.

Furthermore, there are also analog versions of above, with the analog read range being between 0 and 1,023. Also, Arduino supports serial commands such as print, println (i.e., print line), and write. Developers can access information and support for programming Arduino through the online support forum at www.arduino.cc.

Another factor that provides flexibility and advantages to the Arduino IoT device is its ability to integrate sensors and shields. Specifically:

- *Arduino sensors*: They can be both binary or of a range. They measure a range of values, through varying their resistance to reflect their detection. Arduinos sense voltages rather than resistances. Moreover, there are sensors that only vary their resistances based on a voltage divider that provides Arduino a voltage as required.
- *Arduino Shields*: Shields are circuit boards that plug into the top of an Arduino to enhance its functionality. Through shields an Arduino can be enhanced with Ethernet modules, Global Positioning System (GPS) modules, motor modules, prototypes, and more.

By enhancing Arduino with sensors and shields, one can build amazing applications based on reasonable effort and at a rather low cost.

1.8.3.2 Raspberry Pi

Raspberry Pi [16] is another popular device that is used for IoT application development in the scope of both education and enterprise applications. It has been developed by the University of Cambridge. Raspberry Pi is a credit card-sized personal computer (PC), which can be plugged into a TV or monitor as a display device. Apart from IoT education purposes, it is used for programming projects, electronic projects, office applications, high-definition (HD) video playback applications, and more. It costs approximately \$35 to \$100 depending on its extras and the purchased flavor.

Raspberry Pi components include:

- Core components of the PC, including the Raspberry Pi board, a prepared operating system SD card, a universal serial bus (USB) keyboard, a display

(comprising high-definition multimedia interface (HDMI), digital visual interface (DVI), or composite input), and a power supply;

- Extras (i.e., optional components), including USB mouse, a local area network (LAN) cable for Internet connectivity, a powered USB hub, and a case;
- Programming languages, which enable the implementation of applications over the PC. Supported languages and associated environments include C, C++, Java, Scratch, and Ruby, but also any language supported on the ARM6 processor. These languages provide great flexibility in the implementation of the applications.

Like Arduino, the Raspberry board provides opportunities for implementing fantastic applications, both for enterprise use and for fun projects. Three simple applications that illustrate the concept of extending Raspberry Pi with IoT-based functionalities are:

- *A simple programmable networked sensor*: This is a low-cost digital temperature and humidity sensor integrated with Raspberry Pi. It is an open-source project available at GitHub at: <https://github.com/jervine/rpi-temp-humid-monitor>.
- *ThinkBox (IoT on Raspberry Pi)*: This is software already installed and configured on Raspberry to enhance it with IoT development capabilities through a visual UI. It enables graphical creation of new applications from a simple web browser. More information about the ThinkBox project is available at <http://thethingbox.io/>.
- *Raspberry People Counter*: This application is based on motion sensors that are wired to the GPIO pins of Raspberry Pi. This is the development of a people counting application [17].

1.9 Summary

This chapter presented an overview of the IoT computing paradigm and of its main components and technologies. It also described some of the popular reference architectures for IoT systems, which provide the means for integrating IoT technologies in sophisticated applications. This chapter described in more detail some of the basic components of IoT applications such as sensors and Arduino boards. These components can be used for developing IoT simple applications for educational purposes. Real-life IoT applications comprise more sophisticated IoT technologies. The rest of the book dives into the details of more advanced IoT components and technologies starting from WSN. WSNs comprise multiple networked sensors and are the natural next step in our IoT journey, which is continued in Chapter 2.

References

- [1] International Telecommunication Union, “Y.2060: Overview of the Internet of Things,” June 15, 2012, <https://www.itu.int/rec/T-REC-Y.2060-201206-I>.
- [2] Atzori, L., A. Iera, and G. Morabito, “The Internet of Things: A Survey,” *Computer Networks*, Vol. 54, No. 15, 2010, pp. 2787–2805.
- [3] Help Net Security, “Number of Active IoT Devices Expected to Reach 24.1 Billion in 2030,” May 22, 2020, <https://www.helpnetsecurity.com/2020/05/22/active-iot-devices/>.
- [4] Zorzi, M., et al., “From Today’s Intranet of Things to a Future Internet of Things: A Wireless- and Mobility-Related Views,” *IEEE Wireless Communications*, Vol. 17, No. 6, 2010, pp. 44–51.
- [5] Soldatos, J., O. Lazaro, and F. Cavadini, (eds.), *The Digital Shopfloor: Industrial Automation in the Industry 4.0 Era*, Denmark: River Publishers, 2019.
- [6] Boussard, M., et al., “Providing User Support in Web-of-Things Enabled Smart Spaces,” *Proc. of 2nd International Workshop on Web of Things*, Vol. 11, June 2011.
- [7] Guinard, D., et al., “From the Internet of Things to the Web of Things: Resource Oriented Architecture and Best Practices,” in *Architecting the Internet of Things*, New York: Springer, 2011, pp. 97–129.
- [8] Perwej, Y., et al., “An Extended Review on Internet of Things (IoT) and Its Promising Applications,” *Communications on Applied Electronics (CAE)*, Foundation of Computer Science FCS, New York, Vol. 9, No. 26, February 2019, pp. 8–22.
- [9] Da Xu, L., W. He, and S. Li, “Internet of Things in Industries: A Survey,” *IEEE Transactions on Industrial Informatics*, Vol. 10, No. 4, November 2014, pp. 2233–2243.
- [10] International Standardization for Organization, “ISO 7498-1 – Open Systems Interconnection,” November 1994, [http://standards.iso.org/ittf/PubliclyAvailableStandards/s020269_ISO_IEC_7498-1_1994\(E\).zip](http://standards.iso.org/ittf/PubliclyAvailableStandards/s020269_ISO_IEC_7498-1_1994(E).zip).
- [11] Microsoft, “Azure IoT Reference Architecture,” January 9, 2019, <https://docs.microsoft.com/en-us/azure/architecture/reference-architectures/iot/>.
- [12] AWS, “Reference Architecture,” <https://aws.amazon.com/lambda/resources/reference-architectures/>.
- [13] Gubbi, J., et al., “Internet of Things (IoT): A Vision, Architectural Elements, and Future Directions,” *The Computing Research Repository (CoRR)*, July 2012.
- [14] OMDIA/Tractica, “Artificial Intelligence Software Market to Reach \$118.6 Billion in Annual Worldwide Revenue by 2025,” April 30, 2019, <https://tractica.omdia.com/newsroom/press-releases/artificial-intelligence-software-market-to-reach-118-6-billion-in-annual-worldwide-revenue-by-2025/>.
- [15] Arduino, 2020, <https://www.arduino.cc>.
- [16] Raspberry Pi, 2020, <https://www.raspberrypi.org/>.
- [17] Pelaez, A., “Building a People Counter with Raspberry Pi and Ubidots,” August 23, 2013, <http://blog.ubidots.com/building-a-people-counter-with-raspberry-pi-and-ubidots>.